Grade: Check Plus

Problem Analysis

The following probability distribution represents the amount X withdrawn from a convenience store ATM:

Amount x	Probability f(x)
10	$0.10
20	$0.15
40	$0.30
60	$0.15
80	$0.25
100	$0.05

a. Probability of withdrawing at least 60

To find the probability that a customer withdraws at least $60, we sum the probabilities for X=60,80, and 100:

✓ Qa

$P(X \geq 60)=f(60)+f(80)+f(100)$

$P(X \geq 60)=0.15+0.25+0.05=0.45$

b. Expected Value and Standard Deviation

To calculate the mean (μ) and standard deviation (σ), we use the following formulas:

1. Expected Value E(X)

✓ Qb

$E(X)=\sum x \cdot f(x)$

$E(X)=(10 \cdot 0.10)+(20 \cdot 0.15)+(40 \cdot 0.30)+(60 \cdot 0.15)+(80 \cdot 0.25)+(100 \cdot 0.05)$

$E(X)=1+3+12+9+20+5=\$50$

2. Variance Var(X) and Standard Deviation σ

First, calculate $E(X_2)=\sum x_2 \cdot f(x)$:

- $10_2 \cdot 0.10=100 \cdot 0.10=10$
- $20_2 \cdot 0.15=400 \cdot 0.15=60$
- $40_2 \cdot 0.30=1{,}600 \cdot 0.30=480$
- $60_2 \cdot 0.15=3{,}600 \cdot 0.15=540$
- $80_2 \cdot 0.25=6{,}400 \cdot 0.25=1{,}600$
- $100_2 \cdot 0.05=10{,}000 \cdot 0.05=500$
- **Total E(X2)=3,190**

✓ Qc

$Var(X)=E(X_2)-[E(X)]_2=3{,}190-50_2=3{,}190-2{,}500=690$

$\sigma \approx \$26.27$

c. Response to the Manager

I would explain to the manager that the **expected value** is a long-run average, not a value that must be possible for a single transaction. While no individual can withdraw exactly $50 (as the machine only provides specific increments), if we averaged the withdrawals of thousands of customers, that average would gravitate toward $50. It represents the "center of mass" of the probability distribution.

✓ Qd

d. Probability within One Standard Deviation

One standard deviation from the mean is the range $(\mu-\sigma,\mu+\sigma)$:

$50-26.27=23.73$

$50+26.27=76.27$

The possible values of X that fall within the interval [23.73,76.27] are 40 and 60.

$P(23.73 \leq X \leq 76.27)=f(40)+f(60)=0.30+0.15=0.45$

e. Probability within Two Standard Deviations

✓ Qe

Two standard deviations from the mean is the range $(\mu-2\sigma,\mu+2\sigma)$:

$50-2(26.27)=50-52.54=-2.54$

$50+2(26.27)=50+52.54=102.54$

The possible values of X that fall within the interval $[-2.54, 102.54]$ are 10,20,40,60,80, and 100. Since this covers all possible outcomes in the distribution:

$P(-2.54 \leq X \leq 102.54) = 1.00$